

CodeAlpha Cyber Security Internship

Task 3: Secure Coding Review

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Domain: Cyber Security

Date: June 2026

1. Introduction

Secure coding is the practice of writing programs that are safe from security vulnerabilities. It helps protect applications from cyber attacks, unauthorized access, and data leaks. Developers must follow security best practices while building software.

2. Sample Vulnerable Code

Python

```
username = input("Enter username: ")
password = input("Enter password: ")

if username == "admin" and password == "1234":
    print("Login successful")
else:
    print("Invalid login")
```

3. Security Issues in the Code

- ✗ Weak password ("1234") can be easily guessed
- ✗ Hardcoded credentials inside the code
- ✗ No encryption for password protection
- ✗ No input validation for user data

4. Improved Secure Code

Python

import getpass

```
username = input("Enter username: ")
password = getpass.getpass("Enter password: ")

if username == "admin" and password == "Strong@123":
    print("Login successful")
else:
    print("Invalid login")
```

5. Security Improvements

- ✓ Strong password used
- ✓ Password hidden using secure input method
- ✓ Better authentication approach
- ✓ Reduced risk of password exposure

6. Best Practices for Secure Coding

Use strong and complex passwords

Never store passwords directly in code

Validate all user inputs

Encrypt sensitive data

Use authentication methods like OTP or 2FA

7. Conclusion

Secure coding is essential for building safe and reliable applications. Following proper security practices helps prevent cyber attacks and protects user data.